

Lab 1: Microstrip Structures

1.0 Motivation

In this assignment you will review the basics of microstrip transmission lines and a variety of useful structures that can be readily made on FR4 circuit boards with adhesive copper foil tape and BNCs. You will study the use of standard RF measurement equipment, such as the vector network analyzer (VNA) and the time domain reflectometer (TDR), giving you the bench time necessary to build up a proficiency with the instruments.

2.0 Background

Although the textbook and other materials contain numerous equations, here are a few selected ones that you will likely find most useful for the lab exercises.

- The characteristic impedance of a microstrip line of width W separated from a ground plane with a lossless dielectric of thickness H is reasonably well approximated by

$$Z_0 = \sqrt{\frac{L}{C}} = \sqrt{\frac{\mu_0}{\epsilon_r \epsilon_0}} \cdot \frac{H}{W} \cdot \frac{1}{1 + 1.735 \epsilon_r^{-0.0724} \left(\frac{W}{H}\right)^{-0.836}}, \quad (\text{EQ 1})$$

where ϵ_r is the relative dielectric constant of the board material, which generally varies from batch to batch, and with frequency, as mentioned in class and in the text.

- The wavelength of a sinusoidal signal traveling along a microstrip line is

$$\lambda = \frac{\lambda_0}{\sqrt{1 + 0.63 \cdot (\epsilon_r - 1) \cdot \left(\frac{W}{H}\right)^{0.1255}}}, \quad (\text{EQ 2})$$

where λ_0 is the wavelength in vacuum. The formula is most accurate for $W/H > 0.6$. That inequality is reasonably well satisfied for most practical lines.

- The fringing capacitance per width W of a wide open-circuited line is approximately

$$\frac{C}{W} = \epsilon_r \epsilon_0 \cdot \frac{H}{2} = \frac{\epsilon_r \epsilon_0}{2}. \quad (\text{EQ 3})$$

Notice that H ultimately cancels out of the equation.

This approximation is equivalent to extending each edge by an amount equal to $H/2$.

3.0 Experiments

We advise that you carry out the following exercises in the order presented. Document your steps carefully in a lab notebook, and **save the completed circuit boards** for use in future experiments. Your documentation should enable you to explain your designs, present key measurement results (including TDR/VNA plots), and otherwise demonstrate your *quantitative* understanding of the observations including, most importantly, any discrepancies.

3.1 50 Ω lines

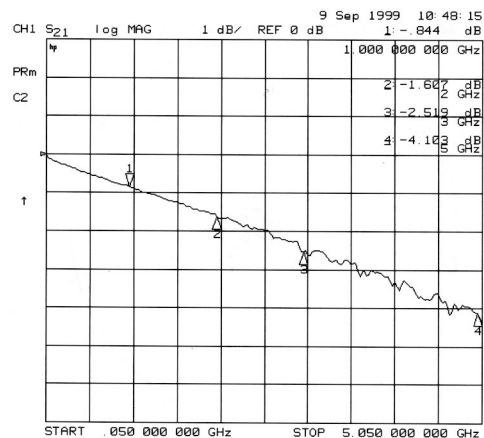
Although in subsequent lab exercises you will have the option to use SMA connectors, here you must use BNCs in order to gain experience with how to compensate for nonidealities. The lessons learned will remain valuable for other connector types. Mount two BNC bulkhead connectors on an FR4 board, as demonstrated during the lab tutorial. You may have to abrade the connector surface with sandpaper before soldering, as BNCs from some manufacturers have a solder-resistant (!) coating on them. Cut copper foil for a 50 Ω line and mount it on the board. Useful information:

- FR4 boards that we use (this week, made by some unknown manufacturer) have a relative dielectric constant ϵ_r around 4.5 at low frequencies (but your mileage may vary). The boards are 1/16" thick (1.6mm) and are plated with 1oz. of copper per square foot (thickness $t = 1.34$ mils, or about 34 μm , usually rounded to 35 μm).
- You will probably discover that $W = 3\text{mm}$ produces about the right Z_0 (49.3 Ω , assuming $\epsilon_r = 4.4$). Copper tape of 1/8" gets you close enough for most purposes. [If we run out of 1/8" width Cu tape, manually slice -- carefully -- 1/4" tape. Care in this step will reap big rewards later on, so try to cut as evenly as you can.]

Evaluate your transmission line using the TDR, with one of the line ends terminated in a 50 Ω BNC load. **Observe and compensate** for the inductance associated with the connector-board transition.

Before you characterize the line **do not forget to calibrate the VNA** (see the attached notes). Measure S_{11} and S_{21} over a broad frequency range. From an S_{21} plot (dB vs. frequency) measure the loss of 50 Ω lines on FR4. Loss is typically ~0.1 dB/inch/GHz (the plot on the right is for a 10" line to exaggerate the loss), but again, your mileage may vary.

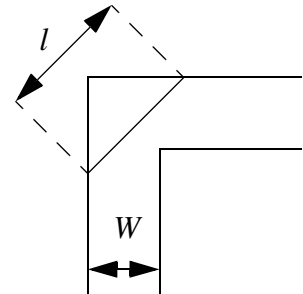
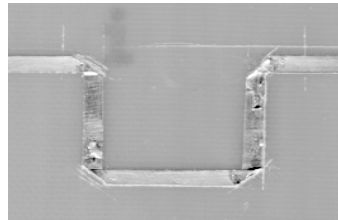
Note that this implies a constant loss of about 0.5dB per *wavelength*. If your plot differs significantly from that shown in the figure, double-check your work. Gross deviations, particularly at higher frequencies, can be a sign of inadequate compensation for the connector-board inductance.



Explore the reflections produced when you load the line with an open and then a short. To short the line you will have to extend the ground plane to the line-side of the board. Constructing a good ground connection is an instructive exercise in its own right. You'll find that it isn't easy (and you should be able to explain why). Making a good ground connection is a valuable skill to acquire.

3.2 90° corners and mitered bends

The world rarely allows you to lay out lines without requiring a turn somewhere. To explore how these behave, construct four consecutive 90° turns as shown in the picture. Using the VNA and/or TDR determine whether the corners look capacitive or inductive.

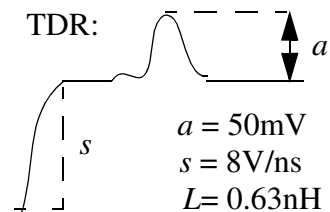
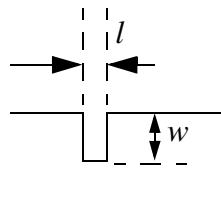
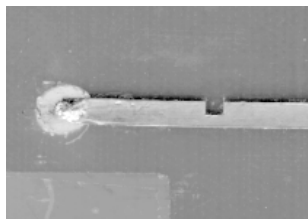


Then, try to minimize impedance discontinuities by trimming the corners as illustrated. Trim the corners one by one, beginning with the one closest to the input, since filtering by a discontinuity reduces your ability to resolve the behavior of downstream discontinuities.

Measure the length, l , of the cut-out section (see diagram). Compare your measured l to the textbook value of $1.8W$.

3.3 Inductive notches

A notch in a transmission line is a convenient way to obtain inductance (if that's your intent). Make a small notch and measure the resulting inductance using the TDR. You will probably observe something like the TDR trace shown here:

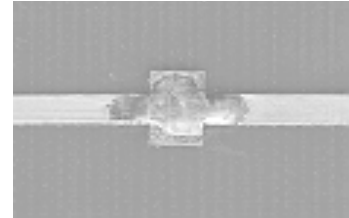


Methods that rely on edge-speed measurements are usually too sensitive to various sources of error, so use the integral method, rather than the derivative method, to estimate reactance. This advice holds for both inductive and capacitive transients. To validate this advice, cross-check your computations using the derivative method. Comment on the likely sources and magnitudes of error for both approaches, and explain why the integral method should yield more reliable values. (Or convincingly refute our advice!)

3.4 Capacitive bulges

Explore the effect of a modest widening in the microstrip line. First estimate the additional capacitance using the parallel plate capacitance formula. Then use the TDR to determine the capacitance experimentally (again, the area method is generally more accurate, but feel free to explore alternatives).

Determine whether a correction for fringing capacitance is needed. Does the capacitance arise from the flanges above and below the line, or from the entire rectangular piece in the center of the picture?



Hint: Do not forget that the capacitor is shunted by *two* 50Ω resistors in parallel when the line is properly terminated. You should be able to explain why this is so.

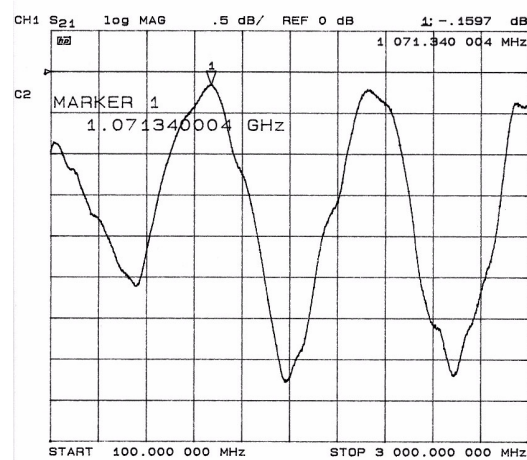
3.5 Halfwave filter

The previous exercise deals with one particular manifestation of a more general effect: A discontinuity can in fact behave as a bandpass filter, passing signals when the length of the discontinuity is an integer multiple of a half-wavelength:

$$\lambda_{pass} = \frac{2l}{m}, m = 1, 2, 3 \dots \quad (\text{EQ 4})$$

The short widening that we examined in the previous section is a bandpass filter for very high frequencies and, as you know, a resonator approximates a pure inductance or capacitance far from center frequency.

The text expands on this idea, and describes how a periodically loaded structure can be used as a lowpass *and* bandpass filter. Construct a 1GHz bandpass filter of this type. Keep in mind that the half-wavelength of a 1GHz signal on FR4 is very roughly of the order of 75-80mm, or about 3", so you don't have space for too many sections on our standard boards. Do your best.



Measure the -3dB bandwidth, maximum stopband rejection, and insertion loss. Explain why the insertion loss is as large as it is.

Although we will study filters in much greater detail later, it should seem intuitively reasonable that the wider the wide section, the greater the mismatch at frequencies where the filter is to reject signals and, hence, the better the filter action. In future labs you will construct a receiver front end, consisting of an antenna, filter and LNA, so try to begin acquiring some insight about filters now.

4.0 Appendix: Network Analyzer Calibration

Depending on how abused a given VNA has been, calibration can be a lot like a visit to Monte Carlo (or any of a number of vending machines around the campus). You should get in the habit of checking calibration quite often (“If you get $S_{21} > 0$ dB for a passive network, it’s time to recalibrate the VNA!”). Inability to hold calibration over a short period of time (e.g., minutes) is most often a sign of inattention to connectors, including pilot error and poor connector maintenance. **A common n00b error is to tighten both halves of an APC-7 connector.** As emphasized in class and in the lab tutorial, **ONLY ONE** half should be tightened. The other half should rotate freely. Think: Tightening both would force the center conductors *away* from each other, resulting in a poor contact.

Calibrate the instrument with the calibration kit, which includes reference shorts, opens, and a 50Ω transmission standard. **Calibration standards and connectors are quite expensive (thousands of dollars!), so please be careful and treat them with respect. In particular, don’t leave them on the benchtops, where they will roll off onto the floor, get dented and dusty, and disappear into oblivion. Put them back in their cases as soon as you are finished with them.**

We were once forced to perform calibrations using only SMA standards (because these are much more common), even though all of the experiments use BNC connectors. Happily we now have BNC standards, one of which is from a trusted vendor, and another that I have made by hand. Be aware that the random BNC 50Ω terminators floating around the lab are NOT to be used in calibration, as they almost always made out of a through-hole resistor with long leads. If, for some reason, the correct BNC 50Ω standard has been misplaced, you may use an N-type 50Ω standard (which we do have) for the particular step in the calibration suite calling for a reflection-free termination. The result is more than good enough over the frequency range that one would use a BNC.

Since the phase of a signal changes as it propagates through the cables/connectors, you might need to perform a mathematical *port extension* if you are using fixturing that extends beyond the plane where the calibrations were performed. Without dialing in a port extension value an open will have $\Gamma = e^{j\beta(f)x}$, where x is the cable length, rather than $\Gamma = 1$. One performs port extensions *after* the main calibration, using the built-in function of the VNA. Large port extension values are to be avoided, because such purely mathematical corrections necessarily fail to take into account the actual physical structure of the setup. During the calibration step, you should include as much of the final setup as possible, and use a post-calibration port extension to perform only the tiniest of tweaks. Typically, port extensions are used to compensate for slight differences in the length of connectors used in the DUT, versus those for the calibration standards. If you are using this capability to compensate for meters of extra cable, etc., STOP. Instead, perform a calibration with those meters of cable in place, then use small port extensions if necessary to fix up what should be 2nd-order remaining discrepancies.

VNA calibration and fixturing

Introduction

What, exactly, goes on in a VNA calibration? How do we do it right? The purpose of this note is to make the process seem less arbitrary, and to give you sufficient understanding to avoid common pitfalls and, on occasion, improvise workarounds in case some calibration standards are unavailable. This last consideration is particularly relevant if using BNC connectors (as we are).

The reference plane: What is it?

As one moves away from the lumped regime of R , L and C , voltage and current along a wire are no longer position-independent. It is then more sensible to reformulate the characterization in terms of the power of incident and reflected *waves*.

For a one-port system, the VNA measures the ratio of reflected to incident power. This ratio is denoted S_{11} , where the first digit of the subscript identifies the port from which the reflected signal emerges, and the second identifies the port into which an incident signal is supplied. For a passive element, the reflection coefficient has less than unit magnitude, and so the dB representation has a negative sign. The *return loss* is the reciprocal of the reflection coefficient. In dB form the return

loss and reflection coefficient will have opposite signs, but the same magnitude. For passive networks, the dB version of return loss will be positive. A large (e.g., 15dB or greater) return loss implies a good match.

The tricky part for a VNA is determining or defining the precise boundary (called the *reference plane*) where the instrument ends and the device under test (DUT) begins, even if there are cables, adaptors and other elements along the path. Calibration consists of connecting a series of impedance standards to the system at the reference plane chosen by the user. If cables are interposed between the VNA and DUT, performing the calibration at the VNA's ports means that the VNA will consider the cables and DUT together as the system to be characterized. If the calibration is performed where the cables connect to the DUT, then the VNA ignores the cables to the best of its ability and thus characterizes only the DUT connected to the end of the cables. The user thus cannot randomly calibrate at arbitrary points. One must always calibrate at the interface that the user has chosen as the reference plane.

In most cases, you will perform the calibrations as if the VNA + cables were the instrument, and everything else were the DUT. In that case, the cables are effectively removed from the measurement. It is good practice to minimize the length of such cables to the extent practical, as cables introduce loss and other artifacts that can degrade the quality of the calibration. The general rule in RF circuits is to keep interconnect as short as possible.

The calibration standards

The VNA needs three impedance standards to carry out calibrated one-port measurements: Open, short, and load (a resistor of Z_0). These three suffice to establish the location of the measurement plane, and to correct VNA errors. Once it calculates a bunch of coefficients, the VNA is in principle ready to make measurements.

For numerous practical reasons, it is best to perform the calibration over the smallest acceptable frequency range. Spreading the always-finite number of correction coefficients over an unnecessarily broad frequency range can degrade accuracy. For example, if you are characterizing a DUT only up to 1GHz, there's no good reason to calibrate the VNA to 20GHz. Some models (and vintages) of VNA are less sensitive to this issue than others, but it is simply good practice not to ask much more of your instrument than you need. Moderation is a good policy here, as in life.

As the foregoing should convey, the quality of the resulting measurements depends critically on the quality of the standards. They are often hand-machined by skilled

artisans, and in relatively low volumes, then characterized and tested extensively. That explains the incredibly high cost of the standards. It is not unusual for a set to cost about as much as an automobile. One should therefore treat these standards with considerable respect. They are not merely ordinary connectors, despite sometimes appearing to be so. Letting them roll off the table and hit the floor is a great way to convert a \$10k cal kit into junk.

Calibration standards in N and SMA connectors are the most common in “serious” RF work. Those in BNCs are comparatively rare, as BNCs were created to serve high-volume, low-cost applications, not high-precision ones. Cal kits for BNCs do exist, but their expense is out of proportion to the low-cost mission for which BNCs were developed. So, in EE251 we use homemade “standards”. Fortunately, these more than suffice up to the 1GHz range that is the domain of this course. Please take care of them, janky though they may be, as they still require some effort to fabricate.

The most challenging of the three BNC standards to homebrew well is the 50Ω termination. High-end cal kits implement the load element with a custom-made resistive disk which, needless to say, is not readily available as a standard catalog item. Various attempts to use surface-mount resistors have been reported, showing that it is possible to make a usable BNC load standard up to 1GHz or so.

There is an alternative, inspired by knowledge that the load standard should produce no reflections, independently of location and connector type. The VNA only uses the load standard to undo its own errors, not to establish the reference plane. So, one could use, say, an N load standard instead of a BNC load standard during calibration. This hybrid method depends, of course, on having a good load standard of some kind. The EE251 lab has cal standards for non-BNC connector types, so use one of those when the VNA asks you to connect the load standard during calibration. Use the BNC open and short standards for the other steps of the calibration procedure. This somewhat unorthodox method works very well, yielding calibrations whose quality is limited largely by the BNC connector’s characteristics.

Two-port calibration

Two-port S -parameter for, say, filters or amplifiers include the forward power gain S_{21} (the power emerging from port 2, due to power entering port 1). The reverse power gain is S_{12} , and port 2’s reflection coefficient is S_{22} .

Calibration involves the same steps as for a one-port measurement, with one addition: A zero-loss line (called a “through” line) needs to be connected between ports

1 and 2 to allow the VNA to correct for internal deviations in gain. The VNA will calculate the difference in loss (or gain) between the DUT and the through, and attribute all the difference to the DUT. For our purposes, an ordinary BNC cable will serve as the “zero”-loss through line. The reason this substitution is acceptable is simple: Microstrip is comparatively lossy, so a BNC cable is, by comparison, an approximately lossless line. Use the shortest practical length of cable for this purpose to minimize the loss of the through. By *practical* is meant short, but not so short that it places bending stress on the VNA connectors.

BEWARE: There may be microstrip-based through lines rattling around in the lab. Do not use them to calibrate the VNA. Comparing your lossy line to an equally lossy line will cause the VNA to report a loss of zero for your line. The through standard must be *much better* than the line whose loss you wish to measure.

Appendix: Mounting BNCs the EE251 way

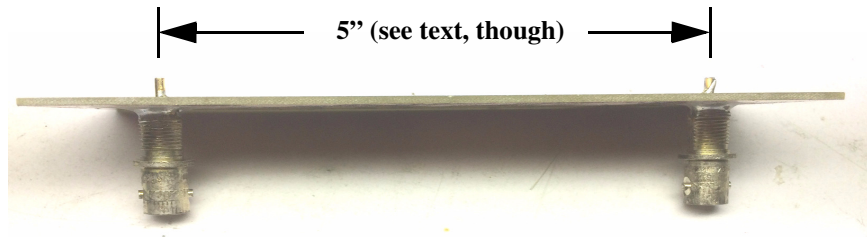
As mentioned in class, our method of mounting BNCs on FR4 is not at all the way one would do so in a product. You will be using the pre-cut 2” x 6” (5.1cm x 15.2cm) rectangular pieces of single-sided FR4 in the lab. That size was chosen to accommodate at least a pair of BNCs separated by 5”, which is (or should be) the spacing between port connectors of the VNAs we have. That allows you to connect your board directly to the VNA without cables if you so desire. That eliminates artifacts associated with lossy or flaky cable.

Note: Not all VNA models have a 5” port-to-port spacing (it’s not a universal standard; not even all HP/Agilent/Keysight models use it), so you should check it with a ruler before drilling holes. And when you do drill, try to be accurate. Although one of the VNA port connectors is designed to permit some lateral movement to accommodate a little slop, it is limited. If you are excessively sloppy and the BNCs do not line up with even a maximally shifted port connector, under no circumstances should you force things. Bending or otherwise damaging the VNA’s connectors will trigger a very expensive repair bill, and cause the instrument to be inaccessible while it’s being repaired. Go back and drill another board instead.

We use “bulkhead” BNC connectors which, as the name suggests, are intended to mount on a chassis and be connected by coaxial cable, rather than on a circuit board with microstrip lines to the connector. Unfortunately, if we were to mount such a BNC in the standard way on our boards, the pin for the center conductor would be far above the surface of the board. So, in order for the microstrip to remain in plane

on the FR4 all the way to the BNC's center conductor, we deliberately drill a hole too small to accommodate the threaded portion of the connector. Instead, we choose a hole size that only accommodates the dielectric. By a miracle, the dielectric protrudes past the threads by the right amount so that its surface ends up flush with the unclad side of the FR4 board.

FIGURE 1. BNCs on FR4 the EE251 way



The hole diameter should be $7/32$ ", the drill bit for which is normally found already in the chuck of the drill press. However, one should verify that the correct bit is indeed in the drill press before proceeding, as you can never know if someone earlier needed to drill a different-size hole and left that bit in the press.

IMPORTANT NOTE 1: If you do change bits, REMEMBER TO REMOVE THE CHUCK KEY before turning on the drill press.

IMPORTANT NOTE 2: FR4 is fiberglass, which cuts like a knife. Do not try to hold the board with your fingers when drilling. If the drill bit grabs the FR4, it will fling it around and cut you, perhaps very badly. Use the clamp to hold the board in place! And make sure that you also use the wood block under your board, or else the drill bit will hit the drill press platform as it breaks through the FR4. That would be bad for both bit and press.

Because we don't use the threaded portion to secure the BNC in the normal way, affix the BNC by soldering the threads to the ground plane. This takes a lot of heat and patience. Just work slowly and deliberately to produce a good, adherent fillet of solder that full encircles the BNC.

Lab 2: Narrow- and Broadband Amplifiers

1.0 Motivation

Having mastered the art of microstrip construction and measurement in Lab 1, you are now ready to tackle the design of active circuits, specifically, two different types of amplifiers. One is a narrowband amplifier that might be suitable as the first stage of a commercial FM broadcast receiver, and the other is a highly useful general-purpose broadband amplifier whose basic topology is found in numerous commercial products.

2.0 Narrowband (single-tuned) amplifier

In this first part of the lab, you'll design a tuned (bandpass) amplifier centered at 98MHz, with a total -3dB bandwidth of at least 20MHz. The corresponding 88-108MHz passband covers the commercial FM radio band. The input and output impedances should provide a sufficiently good match to 50Ω to present a return loss ($|S_{11}|$ and $|S_{22}|$) in excess of 15dB at the center frequency, and in excess of 10dB at the band edges.

To constrain the design space to something manageable, you'll use just one transistor, the 2SC2753 (see datasheets provided in the lab package). Within this constraint, maximize the gain ($|S_{21}|$), above a minimum value of 15dB.

If you elect to wind your own inductor, you might find it useful to recall Harold Wheeler's formula for single-layer cylindrical coils which, in SI units, is

$$L \approx \frac{\mu_0 n^2 \pi r^2}{l + 0.9r} . \quad (\text{EQ 1})$$

As discussed in class, impedance transformations as part of matching may be provided by tapping the inductor appropriately to produce the desired real part. Addition of an appropriate reactive element will remove any remanent reactive part.

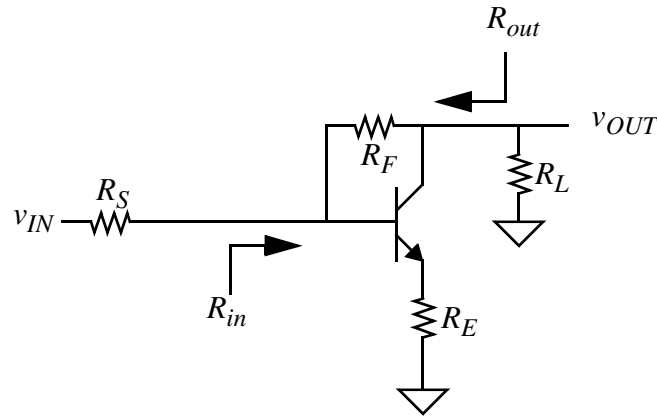
Important reminder: Be sure that your amplifier is AC-coupled at both ports, as discussed in class. Otherwise the instruments could (will) upset your amplifier's bias, and your amplifier might damage the instruments.

3.0 Broadband amplifier

As demonstrated in the previous part of this exercise, a narrowband amplifier is capable of high gain at high frequencies; the gain-bandwidth product "limit" applies to *bandwidth*, not center frequency. For applications that may be served by narrowband circuits, tuned amplifiers provide higher gain (or consume less power for the same gain). They also reduce problems caused by out-of-band sources of interference and noise.

Some applications, though, are truly broadband. This part of the exercise explores the design of a versatile broadband gain block that uses negative feedback to stabilize gain and port impedances (and also reduce distortion at the same time):

FIGURE 1. Shunt-series amplifier (biasing details and DC-blocking capacitors omitted)



As discussed in the book, this topology employs two types of negative feedback together. Emitter degeneration provides one level of negative feedback, which acts with another, provided by R_F . In the usual incarnation of this gain block, the amount of emitter degeneration is sufficient to exert first-order control of the gain (thus stabilizing the gain against variations). The feedback resistance, R_F , then acts primarily to reduce the input and output resistances. If the source and load resistances are equal, it happens that the input and output impedances will also be approximately equal. This property is in addition to the usual improvement in linearity provided by negative feedback. These qualities explain why so many commercial products are based on this topology.

Design generally starts by choosing R_E to set the gain, and then selecting R_F to meet input and output matching objectives. Here, you are to design for a gain of at least 15dB and a return loss of at least 10dB at both ports. The objective is then to maximize the bandwidth over which both sets of specifications are satisfied. You are encouraged to explore shunt-peaking and capacitive bypass ideas to extend bandwidth beyond what would otherwise be the case. Take note of the tradeoffs between achieving good broadband gain and matching.

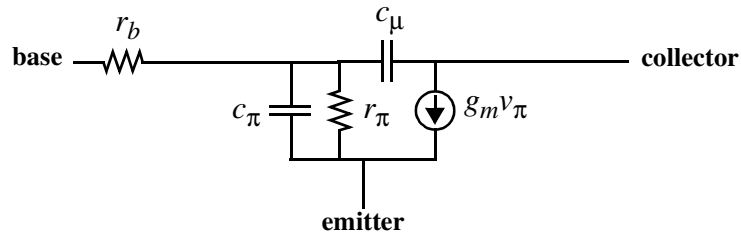
You should provide an open-circuit time constants hand-estimate of bandwidth for the unpeaked amplifier. Consider only the capacitances inherent in the transistor itself (the base-emitter and collector-base capacitances).

As a final recommendation, save even your unsuccessful prototype amplifiers. They may prove useful in providing additional gain when assembling your complete system. Students in prior years have often commented that doing so helps quite a bit.

4.0 Appendix: Summary of bipolar transistor model

The textbook section on bipolar transistor modeling is useful if you haven't had much exposure to the bipolar transistor. Excerpted material is provided here for quick reference:

FIGURE 2. Basic small-signal bipolar transistor model



The model is very similar to that for a MOSFET, as both are fundamentally transconductances (indeed, every active device that has driven the electronics age—vacuum tubes, bipolar transistors, JFETs, MOSFETs, MESFETs, etc.—has been a voltage-controlled current source). However the bias and geometry dependencies of the elements differ a bit. Unlike a MOSFET, a bipolar transistor's transconductance is *solely* a function—and a linear one, at that—of bias (collector) current. At room temperature, a 1mA collector current always produces about a 40mS g_m , independent of the dimensions of the transistor.

Other differences arise in the behavior of the parasitics. The base-to-emitter capacitance results from two different contributions. One is a traditional junction capacitance, and is thus solely junction-voltage dependent. In forward bias, the base-emitter voltage varies little, implying a roughly constant junction capacitance. The other (and generally dominant) component of base-emitter capacitance behaves quite differently; it is linearly proportional to collector *current* and has only a second-order sensitivity to junction voltages.

The bipolar transistor has a resistor, r_π , in shunt with the base-emitter capacitance. This resistance has a value β/g_m . At 1mA collector current and a $\beta = 100$, $r_\pi = 2.5\text{k}\Omega$. At frequencies well above $1/r_\pi c_\pi$, the capacitance c_π dominates the base-emitter impedance.

The collector-base capacitance c_μ is well-modeled as a pure junction capacitance. In normal operation, the collector-base junction is reverse-biased.

The undesired, but unavoidable, extrinsic base resistance r_b is a weak function of bias. At high collector currents, the base current is also large. The voltage thus drops as one moves away from the base contact, assuring that the first moment of the base current distribution moves closer to the base contact as current increases. The consequent reduction in average distance between the contact and the current distribution causes a diminution in r_b . It is not uncommon to see a halving in base resistance under high-level injection conditions. The extrinsic base resistance of the 2SC2753 transistor we use is around 7Ω under nominal conditions, and drops below 5Ω at high currents. Such modest changes are by themselves generally inconsequential except, perhaps, in the design of low-noise amplifiers (about which we'll say more later in the course).

Lab 3: Downconversion Module

1.0 Motivation

By now you have developed familiarity with the design and characterization of active bipolar stages. In this laboratory exercise, you will hone those skills further in the course of designing an oscillator-mixer combination. The output will eventually feed an intermediate-frequency amplifier/filter that you will construct in a future laboratory experiment.

2.0 Mixer

This block accepts the amplified 88-108MHz RF signal and downconverts it to a 10.7MHz (nominal) intermediate frequency. Of the many possible mixer topologies one could imagine employing, you will use a single transistor circuit that exploits the incidental mixing one obtains from a nonlinearity (specifically, the second-order nonlinear term in the series expansion of the i_C - v_{BE} relationship). It is by no means the world's best mixer, but in terms of performance obtained per transistor, it is hard to beat. This type of mixer dates back to the earliest years of the vacuum tube era, and it is still widely used in consumer radios today.

As a suggestion, feed the RF into, say, the base, and the local oscillator (LO) into the emitter circuit. You are certainly free to use other feed arrangements if you wish. Just consider the pros and cons of each option and be prepared to explain the rationale behind your choice. A simple LC tank in the collector circuit should be used to pass only the desired difference-frequency IF component. The bandwidth of this tank just needs to exceed 200kHz in order to pass broadcast FM signals. In this case, we actually want a somewhat large bandwidth to avoid distortion due to nonlinear phase, so a bandwidth of ~300kHz is recommended. There is, of course, a tradeoff between achieving a large enough bandwidth to avoid phase distortion, and having a low enough bandwidth to attenuate the undesired sum-frequency IF component. That's engineering.

3.0 Local Oscillator

You may select any oscillator topology you wish, but just to point you in a direction, consider the Colpitts configuration. It has relatively few failure modes, and well-known fixes for those modes.

We wish to tune over the 88-108MHz band, so your oscillator will also have to tune over a 20MHz range. Theoretically, you have the choice of placing that range above or below the RF range (both will produce a difference frequency equal to the IF). However, placing it above reduces the *percentage* frequency change represented by that fixed 20MHz range. Such “high-side” injection is favored because design difficulty is a strong function of the normalized tuning range you seek. In our case, with a 10.7MHz IF, the LO would need to

tune from 77.3MHz to 97.3MHz (if using low-side injection), or from 98.7MHz to 118.7MHz. The low-side choice obligates us to tune 87.3MHz $\pm$ 11.5%, which implies a total capacitance tuning range of about 46%. High-side injection reduces the tuning requirement to something like a 37% change in capacitance. That difference may not seem dramatic, but you will come to appreciate that the design difficulty is not a linear function of that percentage. [Consider, for example, that making a power converter with 105% efficiency is not a little bit harder than making one with 95% efficiency.]

The MMBV105G/MV105G varactors we stock in the lab (see the collection of datasheets) can accommodate the tuning range requirement without resort to external tuning voltages. We'll make that a requirement: The tuning voltage should be derived from the supply voltage for the rest of the circuit.

4.0 Specs to be achieved

In addition to achieving the tuning range specified, your conversion gain (measured as the IF power delivered by you, divided by the RF power supplied to you) should exceed 10dB. Both ports should exhibit a return loss of at least 10dB (yes, this is much easier than 15dB, by design). It may be difficult with the instrumentation in the laboratory to make this measurement, so please stand by for additional announcements.

Within the constraints of a 12V supply, you are to minimize the supply current consumed to achieve these specifications.

You are to measure the relative suppression of the undesired sum-frequency IF component, compared to the desired difference-frequency component. You need not meet any particular value of this suppression, but you should show that the measured value is explicable from computations carried out on your circuit.

Note: Some students in the past have encountered difficulty by only simulating with values that correspond to an ideal circuit that they did not actually build. If you make this error, you should not be puzzled at the existence of discrepancies! Common errors include assuming that signal sources have zero Thevenin impedance, and failing to take into account various parasitics (e.g., don't treat wires as ideal, impedance-free interconnect -- take the lessons of Lab 1 to heart). Be sure to incorporate knowledge that some parasitics may matter a great deal. Get into the habit of estimating parasitics and including them in your design and simulation methodology.

Lab 4: IF Chain and Demodulator

1.0 Intro

After having engaged in life-and-death combat with the intricacies of single-transistor mixers and oscillators, you are probably ready for a little relaxation. In this penultimate exercise, you will design an IF amplifier that employs a double-tuned filter, followed by an FM demodulator. [In the final exercise of the course—system integration—you will connect everything together and to an antenna and audio amplifier, to see if your radio actually works.]

2.0 IF amplifier

This block is tuned to the same 10.7MHz intermediate frequency produced by the down-converter you built in the previous exercise. To pass the full bandwidth required (as given by Carson's rule, to avoid excessive distortion) you will want to have a 3dB bandwidth of at least 180kHz, and a monotonic (half-) passband. At the same time, the filter needs to attenuate alternate adjacent channels (i.e., the ones 400kHz away) by a “significant” amount. For this exercise, “significant” = 20dB, relative to the center frequency. [It should be relatively straightforward to achieve 25dB rejection and, with care and nontrivial effort, values approaching 30dB should be attainable.] In a commercial FM receiver, multiple stages of such filtering would be employed to provide alternate channel rejections of 50-65dB, typically. We will not be quite so ambitious.

You are to wind your own IF transformer; that will probably be your biggest challenge, time-wise. That will require you to determine what inductances you should target, as well as their mutual coupling. Then you can use the formulas presented earlier to figure out how closely spaced the coils should be. You should not have to iterate too many times to converge on a workable design, because the specifications are sufficiently loose not to require absolute perfection. In lecture, we will review how to determine the coupling coefficient experimentally.

Your IF amplifier *input* should present a >10dB return loss over the passband in order to connect gracefully to your downconverter. The output of the filter should similarly match to 50Ω to that same level of quality, but largely for the purpose of easy, accurate characterization with our equipment. You will almost certainly want to use a transistor at the input and output (e.g., a common-base input transistor, and a follower output transistor), and embed your transformer in between.

The suggested demodulator is a slope detector/leaky peak detector, consisting of a diode, resistor and capacitor. You will need to design this block to filter out the 10.7MHz IF, while not significantly attenuating the baseband modulation (which can go to as high as 15kHz). Given the several orders of magnitude separating those two frequencies, you should not encounter undue difficulty satisfying those requirements.

To avoid loading the filter too much, you can drive this demodulator with the same follower assumed to exist at the output to drive the instrumentation. The effect on S_{22} should not be large. You may, if you wish, also replace the diode with the base-emitter junction of a transistor connected as a follower. This connection implements something that is often called an “infinite impedance” leaky peak detector because it combines the input impedance-boosting properties of a follower with the nonlinearity of a diode.

The combined IF amplifier and filter should provide a minimum of 60dB of insertion gain when terminated in 50Ω . Students in the past advise even greater gain, which is why we advised that you save your leftover amplifier prototypes from previous labs (or borrowed from fellow students). Commercial radio receivers provide overall end-to-end gains well in excess of 100dB, so you see that achieving the bare minimum gains specified will give you a radio that is quite insensitive.

Other hints: Given the impossibility of fabricating perfect coils while simultaneously matching the parasitics and intended loading, achieving correct tuning by construction alone is not a reasonable expectation. Plan on using trimmer capacitors where it makes sense.

If you wind your inductor around some coil form, note that the material’s properties can affect inductance, Q and parasitic capacitance. Plastics like polystyrene or polypropylene are low loss (many pill bottles are made of these plastics). Blue tape is not, as students have discovered experimentally. Also keep the inductor well away from the ground plane (a distance at least equal to the coil diameter is advised). Phenomena such as these argue for doing calculations first so that you know what to expect. Gross discrepancies should trigger investigations directed toward identifying the cause, rather than producing a random flail of randomly changing this and that.

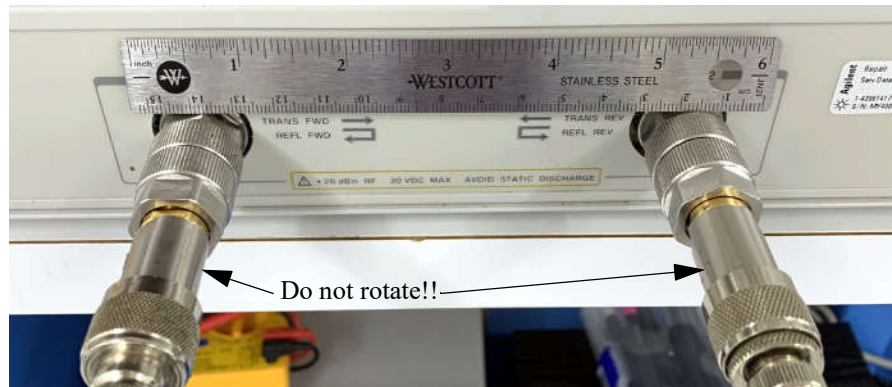
Microstrip, BNCs, VNAs and Mechanical Engineering

As stated in lecture, we deliberately mount the BNCs “wrongly” to provoke non-ideal phenomena that you then compensate.

In general it is advisable to minimize the number of additional transitions (e.g., from connector to cable). Adopting this philosophy minimizes your exposure to bad cables (which regrettably abound), bad connectors, and discontinuities that are ever present when joining this to that.

To facilitate measurements in this class, we advise eliminating cables to connect your 2” x 6” FR4 boards to the VNA by carefully spacing apart the BNC mounting holes as close to 4.8” (about 4-13/16” or 12.2 cm) as you can. That value is the spacing between the ports of the VNAs we have in the lab (see). The manufacturer has allowed for a little “give” in one or both port connections, so if you are a little bit off, the VNA can accommodate it. But do not force things! If your circuit does not easily connect to the VNA because you weren’t quite precise enough in drilling, re-do your board. Or, if that is wholly impractical for some reason, bite the bullet and use cables to plumb things together. Just be sure that you understand the electrical implications of doing so and have a clear idea of how to eliminate or otherwise accommodate the effect of cables.

FIGURE 1. VNA and port spacing is 4.8" (close to 4-13/16")

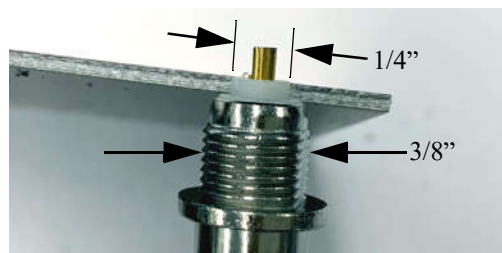


IMPORTANT: When connecting to your board, *NEVER* allow any rotation of the port connectors on the VNA side of things, except for the BNC collar used to secure the VNA to your circuit's BNC. If you fail to observe this rule, you will damage something expensive.

Not all BNCs are identical

The collection of BNCs in the lab is a bit of a mixed bag owing to the long history of this class (including that of its predecessors). The BNCs that you want have a dielectric that extends from the end of the BNC by an amount equal to the thickness of the FR4 boards that we use (1/16", or 1.6 mm or so):

FIGURE 2. BNCs that will work best in EE251



Nearly all of the BNCs we stock should be fine, but you should assume that we've not succeeded 100% at culling. Just hold your connector as shown in the figure and verify that the end of the dielectric will be flush with the non-ground plane side of the FR4 board. If not, please don't put the connector back in the same bin. Put it in some other drawer (your choice) where it won't end up back in circulation as a trap for some unwary student.

The TDR

We are very fortunate to be the beneficiaries of HP/Agilent/Keysight's considerable generosity over many decades. The TDR we have in the lab consists of a 50 GHz-bandwidth oscilloscope mated with an incredibly fast pulse generator ($\sim 7\text{ps}$!). To make such performance possible involves tough engineering tradeoffs, one of which is that the input capacitance of front-end circuits has to be essentially zero, and so they're tiny and thus extremely vulnerable to damage by static charges that other, slower circuits shrug off without a grunt. To prevent killing the front end of this incredible (and incredibly expensive) instrument, you **MUST** don the ESD wrist strap **FIRST**, **BEFORE** you connect your circuit to any part of the instrument. Re-read the previous sentence as many times as necessary to have it become part of your permanent neural net. Then, and only then, plumb things together. After everything's connected power on the TDR/scope and **LAST**, power up your circuit.

FIGURE 3. The TDR, scope, and (blue) wrist strap



Summary

Just carefully follow the rules of the road, and all should be well. We lost one VNA to mistreatment just before I arrived, and none since these rules were explicitly put into print. I hope to keep that record going until I retire, at least!